Deconvolution, and Extended Depth of Field

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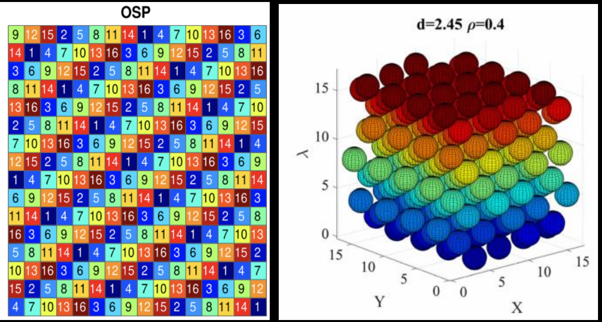
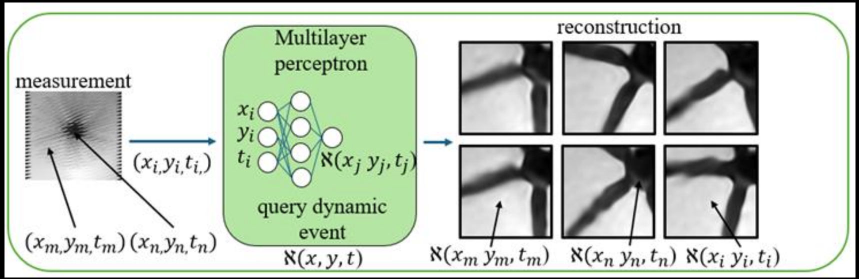
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral

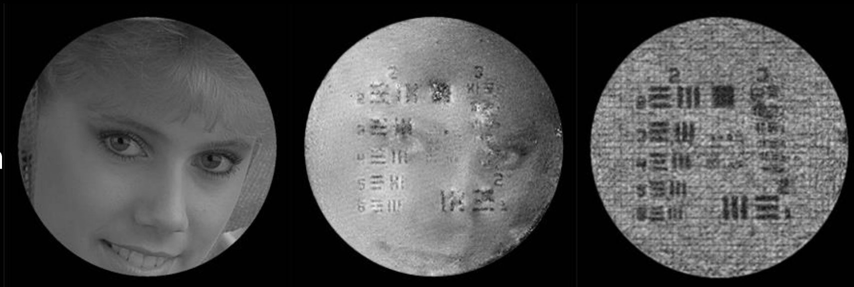


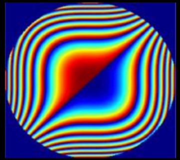
Video



GT phase | Recovered phase | Recovered Amplitude

A woman



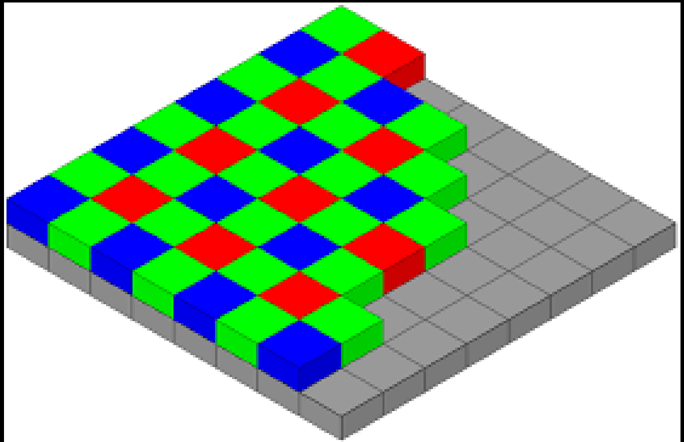


1995

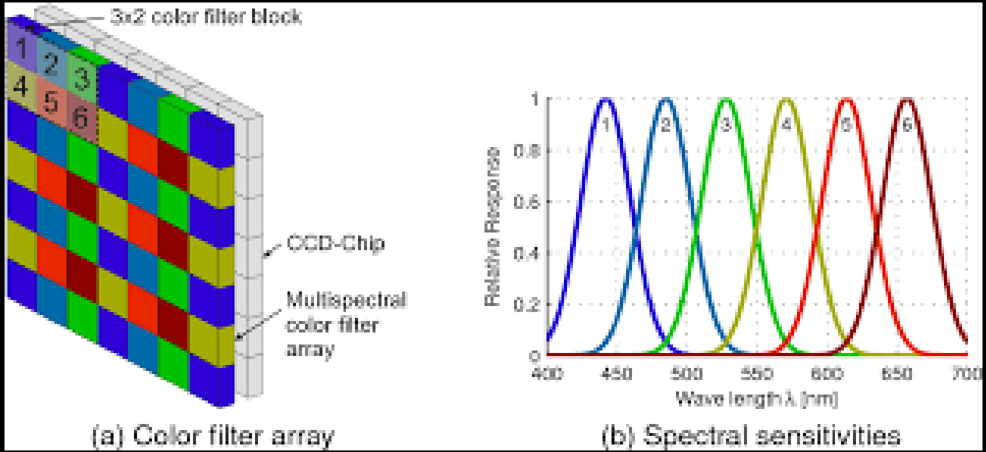
Cubic Phase

Depth

Multispectral Filter Array



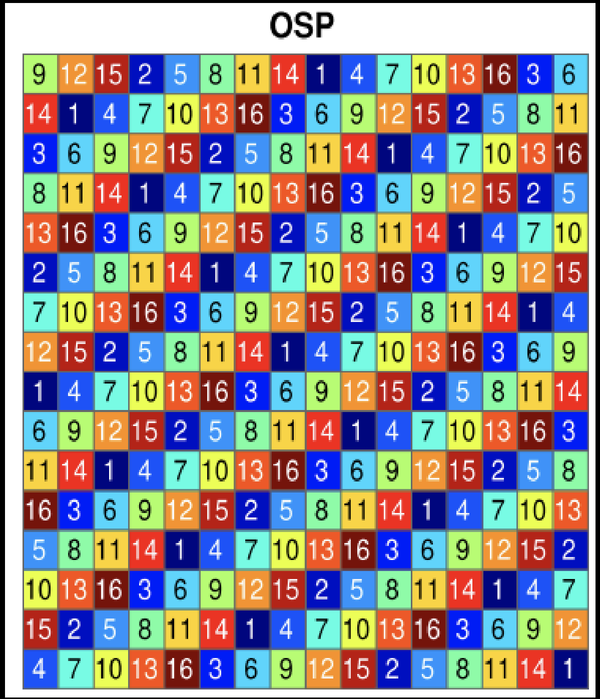
Bayer filter



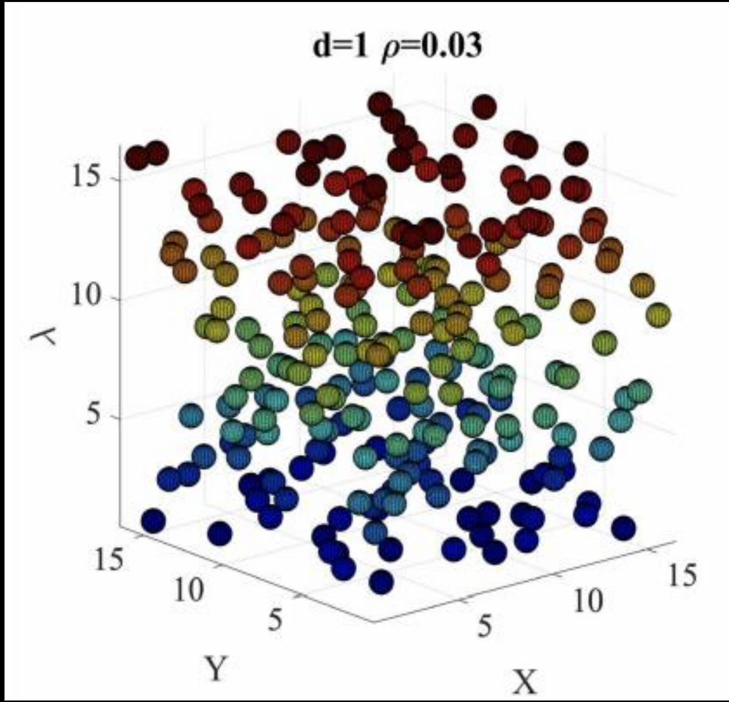
Multispectral Filter Array

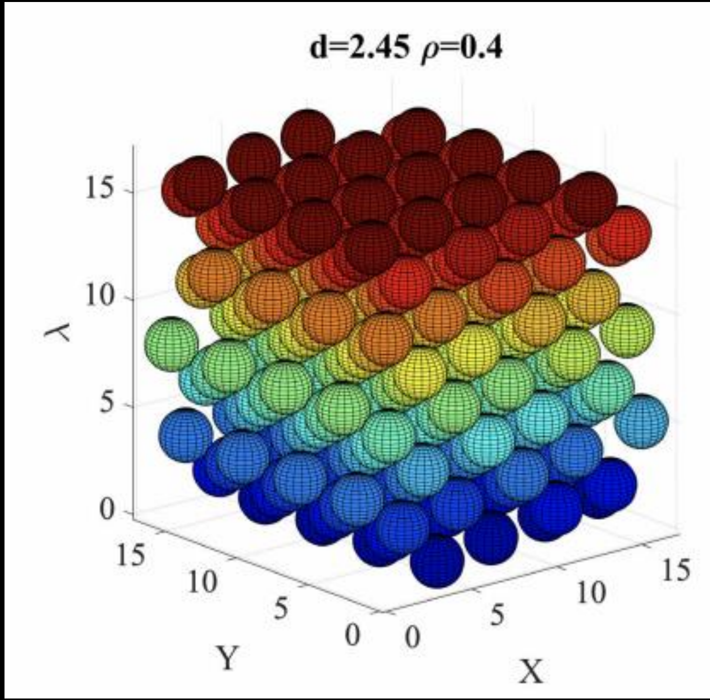
The recovered signal is multispectral datacube

Random vs Sphere Packing

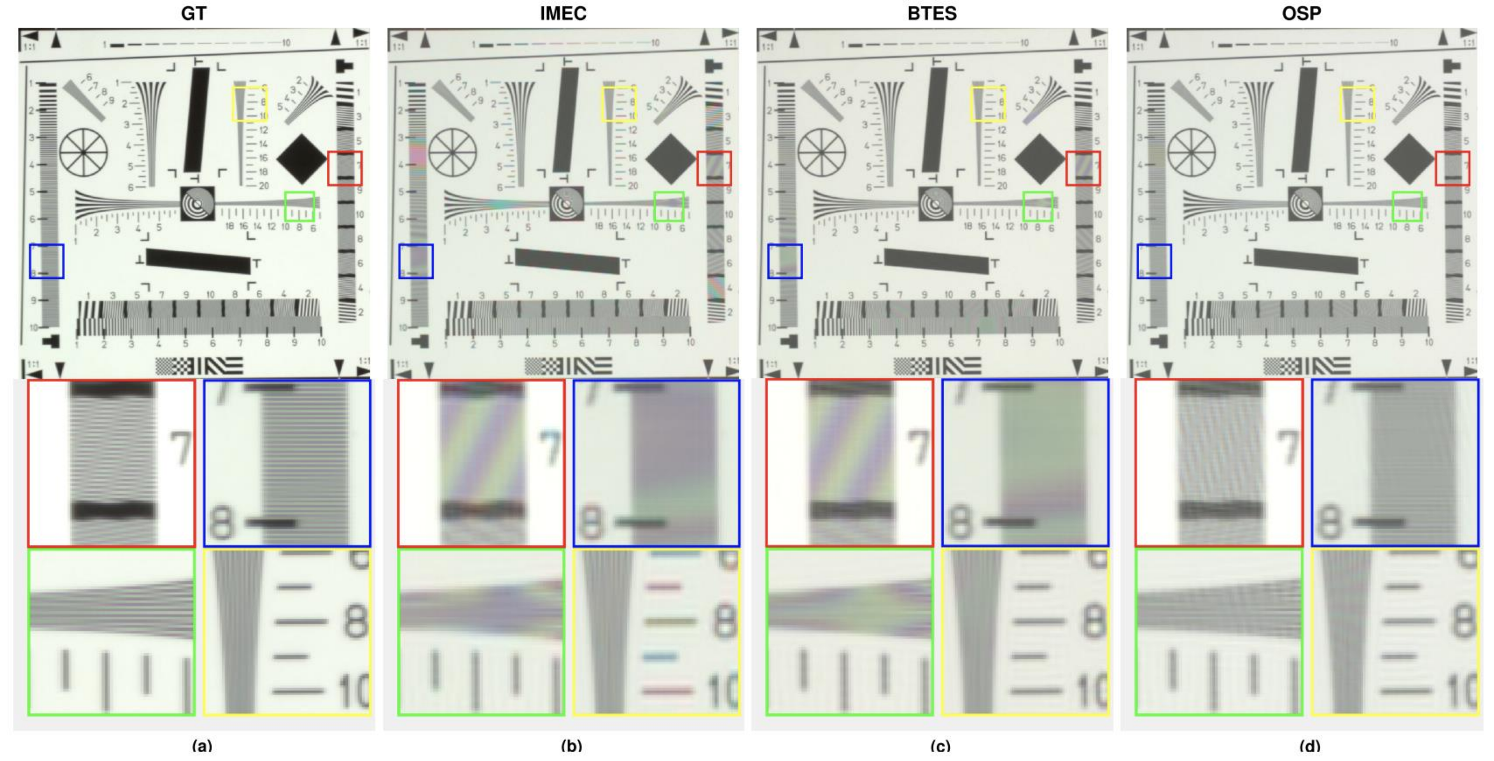


Sphere Packing





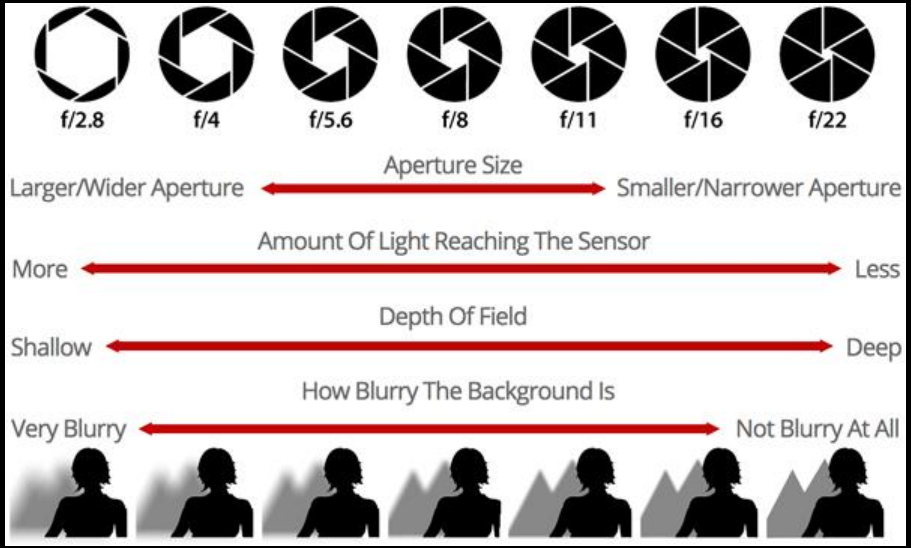
Reconstruction Results using MSFA



N. Diaz, A. Alvarado, P. Meza, F. Guzmán and E. Vera, "Multispectral Filter Array Design by Optimal Sphere Packing," in *IEEE Transactions on Image Processing*, vol. 32, pp. 3634-3649, 2023

Depth-of-Field (DoF)

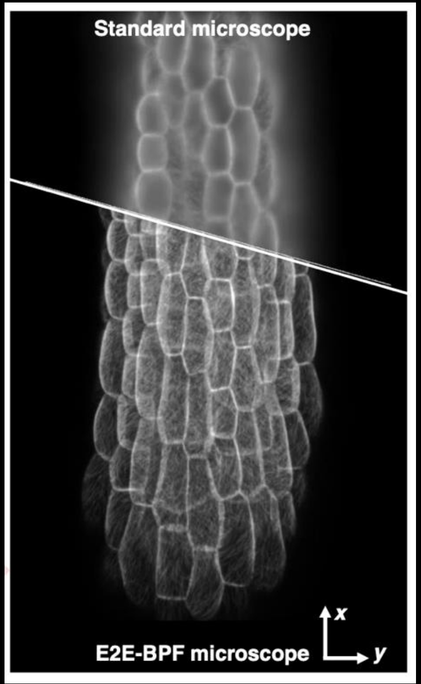
The depth-of-the-field refers to the depth range of an object that can be sharply imaged by a given optical imaging system and is determined by operating wavelength, effective focal length and aperture size of the imaging lens.



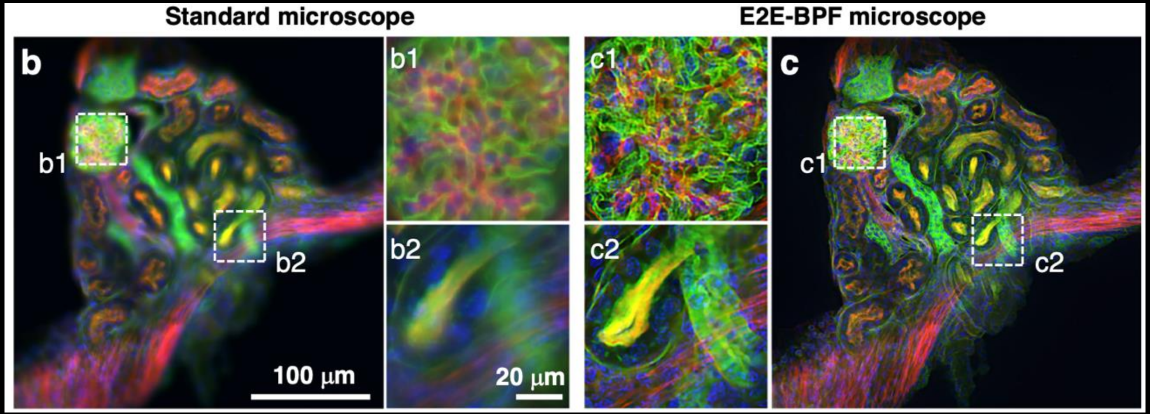
- Reducing the aperture size improves the depth of field
- But reduces the incoming light

Applications of EDoF

High-resolution imaging of large-scale objects without serial refocusing is challenging



E2E-BPF microscope[1]

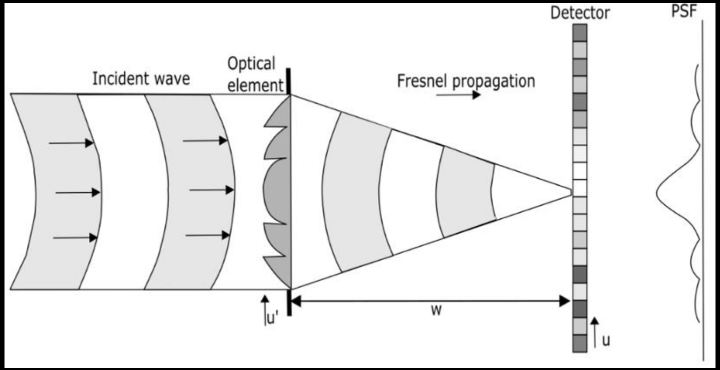


Traditional microscope vs EDoF microscope[2]

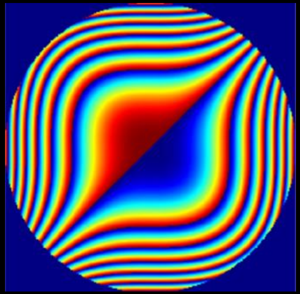
2. Seong, B., Kim, W., Kim, Y. et al. E2E-BPF microscope: extended depth-of-field microscopy using learning-based implementation of binary phase filter and image deconvolution. Light Sci Appl 12, 269 (2023).

Extended-DoF with Diffractive Optical Element

Two assumption when designing EDoF

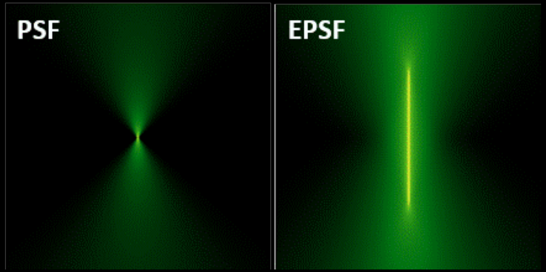


2. It is assumed that any resulting image will be an intermediate image and will require digital processing[1]



1. The incoherent optical system is being modified by a rectangular separable phase mask[1]

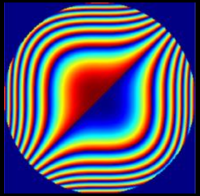
Standard vs Extended PSF



- Conventional cameras has narrow PSF
- The idea of EDoF is to promote PSF insensitive to misfocus

1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

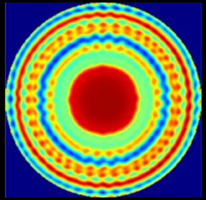
Timeline of EDoF through Wavefront Coding



1995

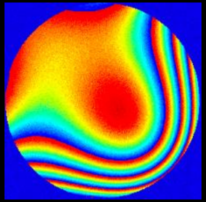
Cubic Phase[1]

$$h_\lambda(s,t) = mod(\alpha\pi(s^3 + t^3), 2\pi),$$
$$|s| < 1, |t| < 1$$



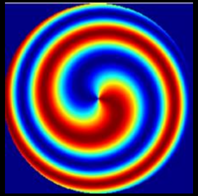
2021

End-to-end optimization[3]



2023

Hardware-in-the-loop[4]



2024

Helical Axicon

$$d(s,t) = e^{\left(jL\theta(s,t)\ +\ j\frac{2\pi}{\lambda}\left(f - \sqrt{f^2 - (s^2+t^2)}\right)\right)}$$

Helical Axicon

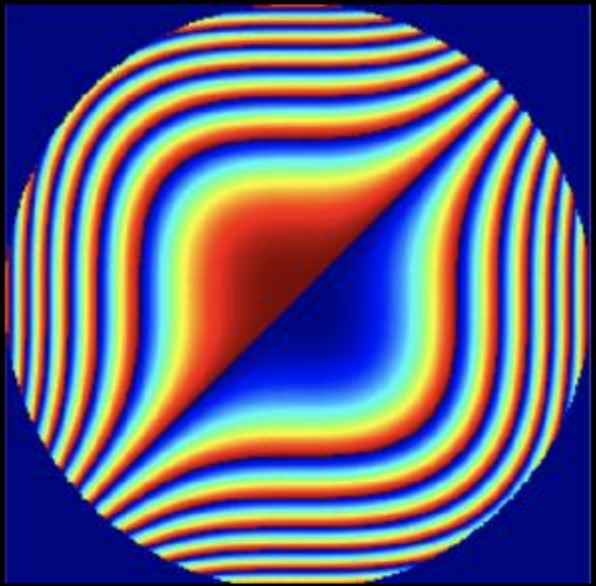
1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).
3. Akpinar, E. Sahin, M. Meem, R. Menon and A. Gotchev, "Learning Wavefront Coding for Extended Depth of Field Imaging," in IEEE Transactions on Image Processing, vol. 30, pp. 3307-3320, 2021.
4. Pinilla, S., Fröch, J. E., Miri Rostami, S. R., Katkovnik, V., Shevkunov, I., Majumdar, A., & Egiazarian, K. (2023). Miniature color camera via flat hybrid meta-optics. Science Advances, 9(21), eadg7297.

EDoF Discrete Model

$$Y_{RGB,z} = S_{RGB}(PSF_{\lambda,z} \circledast X_{\lambda,z}) + \Omega_{\lambda,z}$$

designing the encoding is critical
to facilitate the inverse problem

measurement



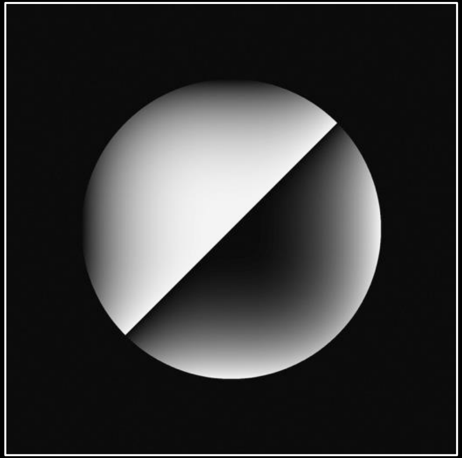
DOE

Scene

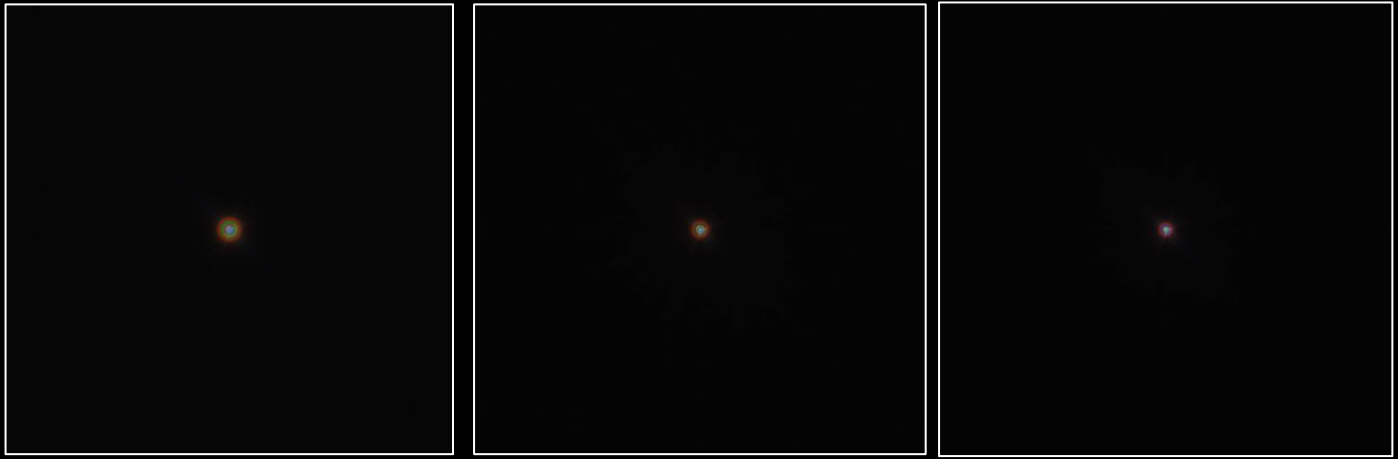
Dowski-Cathey (1995)

$$h_\lambda(s,t) = \ mod(\alpha\pi(s^3 + t^3), 2\pi), |s| < 1, |t| < 1$$

α=[6,16,63,126]



z = 0.5m   z = 1.0m   z = 2.0m



Point spread function at different distances

Dowski-Cathey promotes PSF insensitive to misfocus[1]

1.E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

Mathematical Model

$$PSF_{\lambda,z} = \left| \mathcal{F}\left\{ Q_{\lambda,z}(s,t) \right\}_{\frac{x}{\lambda z_i},\frac{y}{\lambda z_i}} \right|^2,$$

$\mathcal{F}\{.\}$ is the Fourier transform. Where the generalized pupil function is

$$Q_{\lambda,z}(s,t) = A(s,t)e^{\left(j\Phi_\lambda(s,t) + j\Psi_{\lambda,z}\left(\frac{s^2+t^2}{r^2}\right)\right)},$$

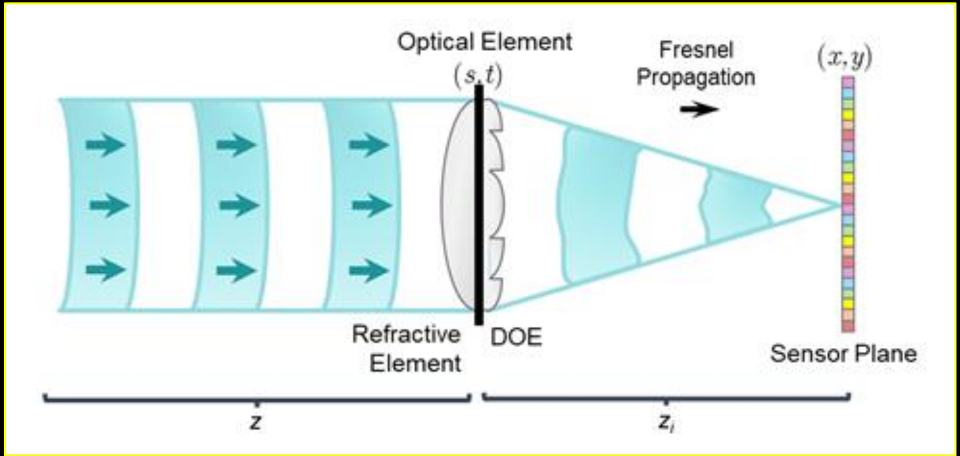
The thickness function of the diffractive element $\Phi_\lambda(s,t) = k(n_\lambda - 1)d(s,t),$

k=2 $\frac{\pi}{\lambda}$ is the wavenumber and n_λ is the wavelength-dependent refractive index.

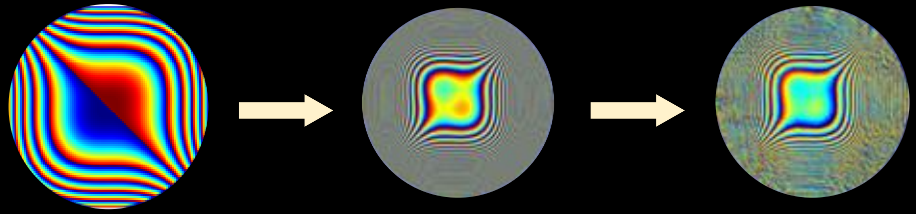
Where the defocus coefficient is $\Psi_{\lambda,z} = \frac{\pi}{\lambda}\left(\frac{1}{z} + \frac{1}{z_i} - \frac{1}{f_\lambda}\right) r^2$

f_λ is the wavelength-dependent focal, r is the radius of the pupil, z distance of the object and zi is the distance from the DOE to the RGB sensor.

Diseño de Difractivos



Inicialización NSGAIII GD



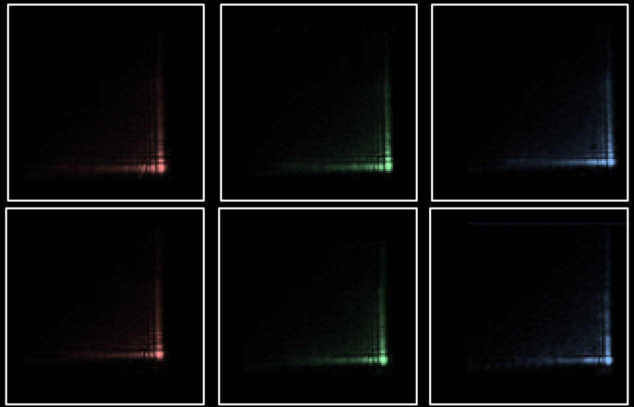
$$\Phi_\lambda^{\mathrm{DOE}} = \Phi_\lambda^{\mathrm{Cubic}} + \Phi_\lambda^{\mathrm{Zernike}} + \Phi_\lambda^{\mathrm{GD}}$$

Sistema Tradicional

Sistema con Difractivo

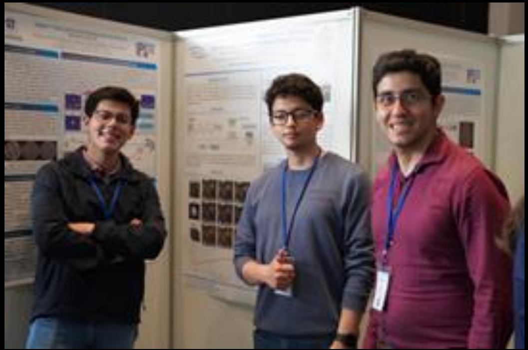
0.5 m

10 m

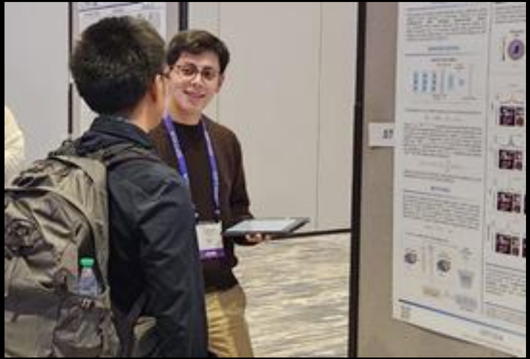


Conferencias y Pasantías

WACCI 2024, Montevideo, Uruguay



Optica Imaging Congress 2025, Seattle, USA



Pasantía IPL UFRO 2025

Conferencias Futuras

International Conference in Image Processing - ICIP 2026

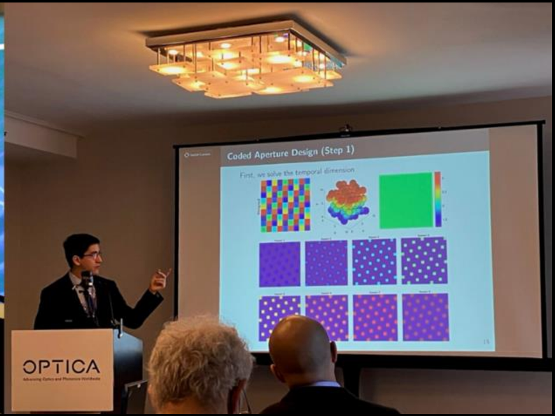
IEEE OJSP

Multispectral Extended Depth-of-Field Imaging via Stochastic Wavefront Optimization

Optica ImageSense Congress 2026

OPTOLAB

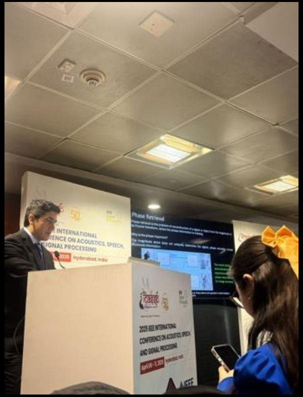
Productivity: Conferences



Optica imaging congress 2023, Boston, USA

2024, IEEE ICASSP, Seul, Corea

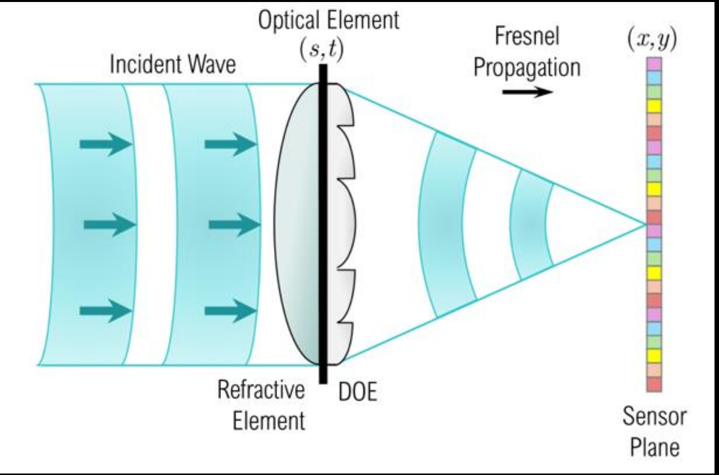
Optica imaging congress 2024, Toulouse, Francia

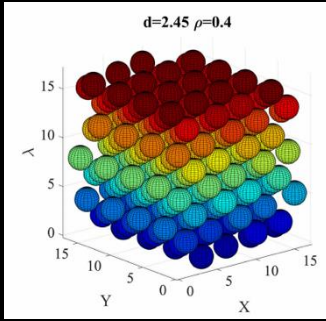


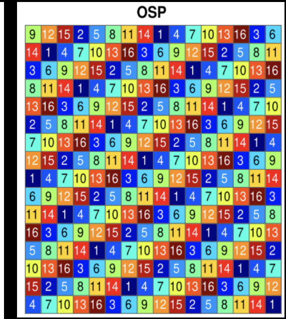
2025, IEEE ICASSP, Hyderabad, India

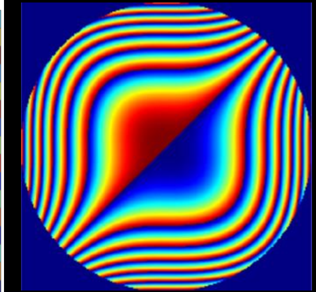
Fondecyt de Iniciación

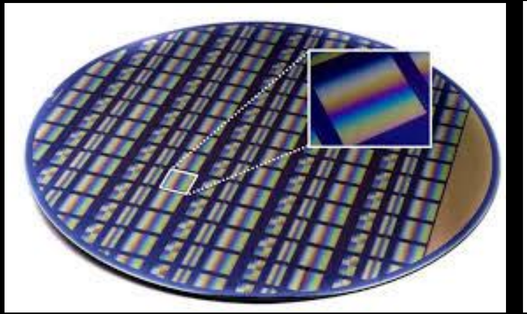
Aim: to design an intelligent, chromatically invariant, encoded wavefront-sensing system for all-in-focus multispectral reconstruction (iCLEAR).











- To fabricate a multispectral camera
- To emmbed an multispectral filter array in camera
- To design a diffractive optical element

Research Network Worldwide

Chile

Esteban Vera
PUCV

Pablo Meza
Universidad de la Frontera

Jorge Tapia
UTFSM

Dario Perez
PUCV

Asia

Felipe Guzmán
University of Tokyo

North America

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Université du Québec, Canada

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Lund University of Lund,
Sweden

Colombia

Henry Arguello, Universidad
Industrial de Santander,
Colombia

Journal Productivity

***IEEE Transactions on Image Processing* (IF 13.7)**

1. **N. Diaz**, A. Alvarado, P. Meza, F. Guzmán and E. Vera, “Multispectral Filter Array Design by Optimal Sphere Packing,” , vol. 32, pp. 3634-3649, 2023.
2. F. Guzman, **N. Diaz**, B. Romero and E. Vera, “Scalable Coding Strategy for High-resolution, High-compression ratio snapshot compressive video,” 2025.

***Optics Express* (IF 3.3)**

1. **N. Diaz**, M Beniwal, M. Marquez, F. Guzman, C. Jiang, J. Liang, and E. Vera, “Single-mask sphere-packing with implicit neural representation reconstruction for ultrahigh-speed imaging,” , 2025.

***Optics & Laser Technology* (IF 5.0)**

1. B. Romero, P. Scherz, **N. Diaz,** A. Cofré, J. Tapia and E. Peters, D Perez and E. Vera, “Phase Retrieval by Designed Hadamard Complementary Coded Apertures,” , pp. 1-13, 2025.

IEEE Open Journal of Signal Processing (IF 2.7)

1. E. Olivia, **N. Diaz**, P. Samuel, and E. Vera, “Multispectral extended depth-of-field imaging via stochastic wavefront optimization,”, pp. 1-13, 2025.

Investigador patrocinante: Dr. Esteban Vera
2023-2026 (Fondecyt Postdoctorado)

Áreas de investigación

- Applied Sphere Packing
- Single snapshot video
- Single Compressive video
- Multispectral Filter Demosaicking
- Phase retrieval with coded Illumination

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